MedTriage Synthetic Training Data Guide

Record format, allowed values, and example

26 October 2026

What one training example represents

Each example is **one patient visit**. The Random Forest predicts the clinician-reviewed final triage level from information available at triage. The training script reads reviewed encounters from the MedTriage SQLite database.

It uses the patient's age, sex, visit type, chief complaint, relevant medical history, symptoms, and vitals as inputs. It uses `final_triage_level` as the target label. A record is included only when `override_status` is `accepted` or `overridden`; pending reviews are skipped.

Sample record


```
{
  "patient": {
    "age": 34,
    "sex": "Female",
    "chronic_conditions": "asthma",
    "allergies": "penicillin"
  },
  "visit_type": "walk_in",
  "chief_complaint": "Fever and headache for three days",
  "symptoms": [
    {
      "name": "High Fever",
      "severity": "moderate",
      "duration": "3 days",
      "notes": "Patient reports chills"
    }
  ],
  "vitals": {
    "temperature": 38.9,
    "blood_pressure": "118/76",
    "heart_rate": 98,
    "oxygen_level": 97,
    "respiratory_rate": 18
  },
  "final_triage_level": "urgent",
  "override_status": "accepted"
}
```


Fields and values

- **Age:** Age in years at the visit. In the database, it is calculated from date of birth and visit date.
- **Sex:** Use the value recorded for the patient, such as `Female` or `Male`.

- **Chronic conditions / allergies:** Text; leave empty or null when none are recorded.
- **Visit type:** `walk_in`, `emergency`, `referral`, or `follow_up`.
- **Chief complaint:** Presenting complaint as recorded at triage.
- **Symptom name:** Select a symptom in the app catalogue, for example `High Fever` or `Persistent Cough`.
- **Symptom severity:** `mild`, `moderate`, or `severe`.
- **Symptom duration:** Text such as `6 hours`, `3 days`, or `2 weeks`.
- **Symptom notes:** Optional additional details; empty/null if none.
- **Temperature:** Degrees Celsius, or null if not measured.
- **Blood pressure:** Text such as `118/76`, or null if not measured.
- **Heart rate:** Beats per minute, or null if not measured.
- **Oxygen level:** Oxygen saturation percentage, or null if not measured.
- **Respiratory rate:** Breaths per minute, or null if not measured.
- **Final triage level:** Reviewed target label: `immediate`, `emergent`, `urgent`, `less_urgent`, or `non_urgent`.
- **Override status:** `accepted` or `overridden` for reviewed records; `pending` records are excluded.

Triage level labels

Score	Label	Project description
1	`immediate`	Immediate attention for life-threatening signs.
2	`emergent`	Very urgent; the project prompt says see within 10 minutes.
3	`urgent`	Urgent; the project prompt says see within 30 minutes.
4	`less_urgent`	Less urgent; the project prompt says see within 60 minutes.
5	`non_urgent`	Routine care according to the project scale.

Building a useful synthetic set

- Include examples from **all five levels** and avoid leaving a level with zero records.
- Vary symptoms, complaints, vitals, severity, duration, and patient context in coherent combinations.
- Use only values that would be known at triage as model inputs. Keep the clinician's final level as the target.
- Do not use the current AI prediction to create the target label; that would train the new model to copy the existing one.
- Do not include patient names, email addresses, or other identifying information.
- Synthetic records can exercise the training workflow, but scores on synthetic data are not evidence of clinical performance.

The training script's default minimum is 50 reviewed encounters. To run it on the smaller seeded demo dataset for a workflow smoke test, from `server/` run:

```
python train_random_forest.py --min-rows 20
```